

CER-AI PREOPERATIVE ECTASIA RISK ASSESSMENT

Corneal Ectasia Risk Assessment Intelligence

The final surgical decision and all associated responsibility and liability rest with the surgeon. This application is a clinical decision-support aid only.

Patient	Sample 03	Age	23
Assessment date	Removed	Reviewer	Removed
Eyes	OD; OS	Overall disposition	STOP-DEFER

STOP-DEFER — do not proceed with elective corneal refractive surgery.

OD — STOP-DEFER

Randleman / ERSS

Component	Points / disposition	Exact finding
Topography	1	
Residual stromal bed	0	
Age	0	
Preoperative pachymetry	2	
Manifest MRSE	0	
Total	3	
Topography category	ASYMMETRIC_BOWTIE	
Disposition	CAUTION	
Decision trigger	CAUTION	ERSS total: 3; topography: ASYMMETRIC_BOWTIE (1 points); signed I-S: 0.67 D; additional scored components: pachymetry: 2 points

NICE

Component	Input	Points
K2	47.9	3
central_pachymetry	495	3
B_Ele_Th	6	1
I_S	0.67	1
Total		8
Classification	CAUTION	CAUTION
Decision trigger	CAUTION	NICE total: 8

PS3

Factor	Status	Exact finding
anterior_km	NORMAL	Anterior Km 47.1 D < 48 D.
thinnest	MODERATE	Thinnest 492 µm is 470-499 µm.
elevation	NORMAL	F.Ele.Th 4 µm <=12 and B.Ele.Th 6 µm <=15.
inter_eye_asymmetry	NORMAL	Inter-eye score 0/5.
srax	NORMAL	Front-map SRAX 2.7° <= 20°. Source: Axial/Sagittal Curvature (Front).
ppi_average	MODERATE	PPI Average 1.37 > 1.20.
corneal_thickness_map_morphology	NOT_EVALUATED	Corneal Thickness Map morphology: not evaluated; surgeon review required; not counted in automated PS3.
relative_thickness_map	NOT_EVALUATED	Relative Thickness Map: not evaluated; surgeon review required; not counted in automated PS3.
pti_ctsp_morphology	NOT_EVALUATED	PTI/CTSP thickness-profile morphology: not evaluated; surgeon review required; not counted in automated PS3.
Moderate / High	2	0
Procedure disposition	STOP-DEFER	Raw PS3 PRK disposition: DEFER. Triggering PS3 finding(s): thinnest: Thinnest 492 µm is 470-499 µm.; ppi_average: PPI Average 1.37 > 1.20.
Decision trigger	STOP-DEFER	Raw PS3 PRK disposition: DEFER. Triggering PS3 finding(s): thinnest: Thinnest 492 µm is 470-499 µm.; ppi_average: PPI Average 1.37 > 1.20.

Astigmatic disparity validation

Status	Magnitude difference	Axis difference
VALIDATION_REQUIRED	0.5	20.7
Interpretation	Astigmatic disparity (magnitude difference 0.50 D; axis difference 20.7°) requires measurement/refraction validation; it is not a PS3 risk factor and does not independently defer LASIK.	Non-scoring; no independent procedure restriction

Belin/Ambrósio BAD-D

Parameter	Value	Interpretation
Final BAD-D	2.09	SUSPICIOUS / CAUTION
Df	1.47	NORMAL / < 1.60; information only
Db	0.07	NORMAL / < 1.60; information only
Dp	3.17	ABNORMAL / >= 2.60; information only
Dt	1.43	NORMAL / < 1.60; information only
Da	1.83	SUSPICIOUS / 1.60 to < 2.60; information only
PPI Min	0.76	NORMAL / < 0.80; information only
PPI Avg	1.37	ABNORMAL / > 1.17; information only
PPI Max	1.71	ABNORMAL / > 1.52; information only
ARTmax	288	ABNORMAL / < 357 µm; information only
Decision trigger	CAUTION	Final BAD-D: SUSPICIOUS

Procedural safety

Parameter	Canonical result	Exact finding
LASIK_PTA_percent	Not documented	
LASIK_RSB_um	Not documented	
PRK_PTA_percent	22.358	
PRK_RST_um	382	
estimated_final_Kmean_D	43.9	
final_Kmean_model	MYOPIC_0.8_D_PER_D_ESTIMATE	
hard_stops	final_kmean: No; lasik_pta: No; lasik_rsb: No; preop_thickness: No; prk_pta: No; prk_rst: No; sphere_magnitude: No	
hyperopic_review_level	Not documented	
hyperopic_treatment_component_D	-3.5	
missing	Not documented	
scalar_final_Kmean_mode_l_valid	Yes	
status	PASS	

Procedure planning

Planning item	Canonical result
OD — STOP-DEFER — PRK	OZ 6.5 mm — TZ 9 mm — Ablation 60 µm
MMC guidance	MANDATORY
rejection_reasons	Not documented
Selected LASIK plan	Not documented
sequence	Not documented

Decision basis

Level	Canonical decision driver
STOP	detail: Raw PS3 PRK disposition: DEFER. Triggering PS3 finding(s): thinnest: Thinnest 492 µm is 470-499 µm.; ppi_average: PPI Average 1.37 > 1.20.; key: ps3; status: STOP-DEFER
CAUTION	detail: ERSS total: 3; topography: ASYMMETRIC_BOWTIE (1 points); signed I-S: 0.67 D; additional scored components: pachymetry: 2 points; key: randleman_erss; status: CAUTION
CAUTION	detail: Final BAD-D: SUSPICIOUS; key: bad_d; status: CAUTION
CAUTION	detail: NICE total: 8; key: nice; status: CAUTION

Canonical Pentacam values and provenance

Canonical field	Value	Provenance
ARTmax	288	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
Final BAD-D	2.09	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
B.Ele.Th	6	BELIN_AMBROSIO_LOWER_ELEVATION_LABEL_VALUE_ROW / De-identified source image
CKI	1.01	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Da	1.83	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Db	0.07	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Df	1.47	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Dp	3.17	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Dt	1.43	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
F.Ele.Th	4	BELIN_AMBROSIO_LOWER_ELEVATION_LABEL_VALUE_ROW / De-identified source image
IHA	12.2	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
IHD	0.013	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
ISV	19	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
IVA	0.11	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Signed I-S	0.67	SURGEON_CONFIRMED
K1	46.4	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
K1 axis	80.7	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
K2	47.9	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
K2 axis	80.7	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
KI	1.04	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
KISA	3.748	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Kmax (Front)	48.4	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
Km	47.1	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
PPI Avg	1.37	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
PPI Max	1.71	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image

Canonical field	Value	Provenance
PPI Min	0.76	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
PS3 BAD Axis (flat meridian, beside K1)	170.7	BELIN_AMBROSIO_CENTER_NUMERIC_BOX / De-identified source image
Pupil Center pachymetry	495	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
HWTW	11.5	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
Thinnest pachymetry	492	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
SRAX	2.7	AXIAL_SAGITTAL_CURVATURE_FRONT_GEOMETRIC / De-identified source image
Astigmatism	1.5	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
Displayed steep/astigmatic axis	80.7	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
Topometric RMin	6.97	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image

Surgeon-completed values

Field	Original	Surgeon-entered value	Provenance
I_S	Not documented	0.67	SURGEON_CONFIRMED

Version provenance

Layer	Version
clinical_policy	CER-AI-2026-10-05-SIGNED-IS-NEGATIVE-BOUNDARY-V13
deployment_sha	f4b2950cb7492664458047cd528684a1a5e2d597
software	2.0.1
source_registry	2026-09-10-ML7-REUSE-CANONICAL-K
srax_algorithm	FRONT_MAP_SRAX_GT20-2026-09-07

OS — STOP-DEFER

Randleman / ERSS

Component	Points / disposition	Exact finding
Topography	1	
Residual stromal bed	0	
Age	0	
Preoperative pachymetry	2	
Manifest MRSE	0	
Total	3	
Topography category	ASYMMETRIC_BOWTIE	
Disposition	CAUTION	
Decision trigger	CAUTION	ERSS total: 3; topography: ASYMMETRIC_BOWTIE (1 points); signed I-S: 0.81 D; additional scored components: pachymetry: 2 points

NICE

Component	Input	Points
K2	48.5	3
central_pachymetry	494	3
B_Ele_Th	6	1
I_S	0.81	1
Total		8
Classification	CAUTION	CAUTION
Decision trigger	CAUTION	NICE total: 8

PS3

Factor	Status	Exact finding
anterior_km	NORMAL	Anterior Km 47.4 D < 48 D.
thinnest	MODERATE	Thinnest 489 µm is 470-499 µm.
elevation	NORMAL	F.Ele.Th 4 µm <=12 and B.Ele.Th 6 µm <=15.
inter_eye_asymmetry	NORMAL	Inter-eye score 0/5.
srax	NORMAL	Front-map SRAX 6.1° <= 20°. Source: Axial/Sagittal Curvature (Front).
ppi_average	MODERATE	PPI Average 1.36 > 1.20.
corneal_thickness_map_morphology	NOT_EVALUATED	Corneal Thickness Map morphology: not evaluated; surgeon review required; not counted in automated PS3.
relative_thickness_map	NOT_EVALUATED	Relative Thickness Map: not evaluated; surgeon review required; not counted in automated PS3.
pti_ctsp_morphology	NOT_EVALUATED	PTI/CTSP thickness-profile morphology: not evaluated; surgeon review required; not counted in automated PS3.
Moderate / High	2	0
Procedure disposition	STOP-DEFER	Raw PS3 PRK disposition: DEFER. Triggering PS3 finding(s): thinnest: Thinnest 489 µm is 470-499 µm.; ppi_average: PPI Average 1.36 > 1.20.
Decision trigger	STOP-DEFER	Raw PS3 PRK disposition: DEFER. Triggering PS3 finding(s): thinnest: Thinnest 489 µm is 470-499 µm.; ppi_average: PPI Average 1.36 > 1.20.

Astigmatic disparity validation

Status	Magnitude difference	Axis difference
NORMAL	0.35	8.2
Interpretation	Astigmatic disparity within validation thresholds: magnitude difference 0.35 D; axis difference 8.2°; no PS3 consequence.	Non-scoring; no independent procedure restriction

Belin/Ambrósio BAD-D

Parameter	Value	Interpretation
Final BAD-D	1.94	SUSPICIOUS / CAUTION
Df	1.17	NORMAL / < 1.60; information only
Db	-0.81	NORMAL / < 1.60; information only
Dp	3.06	ABNORMAL / >= 2.60; information only
Dt	1.53	NORMAL / < 1.60; information only
Da	2.06	SUSPICIOUS / 1.60 to < 2.60; information only
PPI Min	0.71	NORMAL / < 0.80; information only
PPI Avg	1.36	ABNORMAL / > 1.17; information only
PPI Max	1.87	ABNORMAL / > 1.52; information only
ARTmax	262	ABNORMAL / < 357 µm; information only
Decision trigger	CAUTION	Final BAD-D: SUSPICIOUS

Procedural safety

Parameter	Canonical result	Exact finding
LASIK_PTA_percent	Not documented	
LASIK_RSB_um	Not documented	
PRK_PTA_percent	28.246	
PRK_RST_um	350.875	
estimated_final_Kmean_D	42.7	
final_Kmean_model	MYOPIC_0.8_D_PER_D_ESTIMATE	
hard_stops	final_kmean: No; lasik_pta: No; lasik_rsb: No; preop_thickness: No; prk_pta: No; prk_rst: No; sphere_magnitude: No	
hyperopic_review_level	Not documented	
hyperopic_treatment_component_D	-5	
missing	Not documented	
scalar_final_Kmean_mode_l_valid	Yes	
status	PASS	

Procedure planning

Planning item	Canonical result
OS — STOP-DEFER — PRK	OZ 6.5 mm — TZ 9 mm — Ablation 88.125 µm
MMC guidance	MANDATORY
rejection_reasons	Not documented
Selected LASIK plan	Not documented
sequence	Not documented

Decision basis

Level	Canonical decision driver
STOP	detail: Raw PS3 PRK disposition: DEFER. Triggering PS3 finding(s): thinnest: Thinnest 489 µm is 470-499 µm.; ppi_average: PPI Average 1.36 > 1.20.; key: ps3; status: STOP-DEFER
CAUTION	detail: ERSS total: 3; topography: ASYMMETRIC_BOWTIE (1 points); signed I-S: 0.81 D; additional scored components: pachymetry: 2 points; key: randleman_erss; status: CAUTION
CAUTION	detail: Final BAD-D: SUSPICIOUS; key: bad_d; status: CAUTION
CAUTION	detail: NICE total: 8; key: nice; status: CAUTION

Canonical Pentacam values and provenance

Canonical field	Value	Provenance
ARTmax	262	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
Final BAD-D	1.94	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
B.Ele.Th	6	BELIN_AMBROSIO_LOWER_ELEVATION_LABEL_VALUE_ROW / De-identified source image
CKI	1.01	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Da	2.06	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Db	-0.81	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Df	1.17	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Dp	3.06	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
Dt	1.53	BELIN_AMBROSIO_BOTTOM_BAD_D_STRIP / De-identified source image
F.Ele.Th	4	BELIN_AMBROSIO_LOWER_ELEVATION_LABEL_VALUE_ROW / De-identified source image
IHA	9.1	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
IHD	0.01	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
ISV	23	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
IVA	0.1	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Signed I-S	0.81	SURGEON_CONFIRMED
K1	46.4	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
K1 axis	91.8	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
K2	48.5	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
K2 axis	91.8	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
KI	1.04	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
KISA	0.689	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image
Kmax (Front)	49.2	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
Km	47.4	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
PPI Avg	1.36	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
PPI Max	1.87	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image

Canonical field	Value	Provenance
PPI Min	0.71	BELIN_AMBROSIO_PROGRESSION_INDEX_BOX / De-identified source image
PS3 BAD Axis (flat meridian, beside K1)	1.8	BELIN_AMBROSIO_CENTER_NUMERIC_BOX / De-identified source image
Pupil Center pachymetry	494	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
HWTW	11.5	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
Thinnest pachymetry	489	FOUR_MAPS_REFRACTIVE_LOWER_LEFT_LABELED_BOX / De-identified source image
SRAX	6.1	AXIAL_SAGITTAL_CURVATURE_FRONT_GEOMETRIC / De-identified source image
Astigmatism	2.1	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
Displayed steep/astigmatic axis	91.8	SHOW_2_EXAMS_TOPOMETRIC_CORNEA_FRONT / De-identified source image
Topometric RMin	6.86	SHOW_2_EXAMS_TOPOMETRIC_CENTER_INDICES_8MM / De-identified source image

Surgeon-completed values

Field	Original	Surgeon-entered value	Provenance
I_S	Not documented	0.81	SURGEON_CONFIRMED

Version provenance

Layer	Version
clinical_policy	CER-AI-2026-10-05-SIGNED-IS-NEGATIVE-BOUNDARY-V13
deployment_sha	f4b2950cb7492664458047cd528684a1a5e2d597
software	2.0.1
source_registry	2026-09-10-ML7-REUSE-CANONICAL-K
srax_algorithm	FRONT_MAP_SRAX_GT20-2026-09-07